

Statistical and Machine Learning — Revision Class

Teacher Version (with Model Answers)

Contents

1	Block A: Linear Models	2
1.1	True / False	2
1.2	Short Conceptual Questions	2
1.3	Worked Example (Q4)	3
2	Block B: Model Evaluation	4
2.1	True / False	4
2.2	Short Conceptual Questions	4
2.3	Worked Example (Q7)	5
3	Block C: Regularisation and Bayesian Methods	6
3.1	True / False	6
3.2	Short Conceptual Questions	6
3.3	Worked Example (Q10)	7
4	Block D: Kernel and Exemplar Methods	8
4.1	True / False	8
4.2	Short Conceptual Questions	8
4.3	Worked Example	8
5	Block E: Tree-Based Methods and Boosting	10
5.1	True / False	10
5.2	Short Conceptual Questions	10
5.3	Worked Example	10
6	Open Synthesis Questions	12

Teacher notes. Model answers follow each question. The three synthesis questions at the end are optional stretch material.

Suggested 90-minute schedule

Block	Topic	Group work	Discussion	Total
A	Linear Models	12 min	8 min	20 min
B	Model Evaluation	12 min	8 min	20 min
C	Regularisation & Bayes	12 min	8 min	20 min
D	Kernel & Exemplar Methods	8 min	7 min	15 min
E	Tree-Based Methods & Boosting	8 min	7 min	15 min
S	Synthesis (optional)	—	—	overflow

1 Block A: Linear Models

1.1 True / False

#	Statement	Answer	Notes
1	OLS minimises $\ \mathbf{Y} - \mathbf{X}\beta\ ^2$.	TRUE	Definition.
2	$\hat{\beta}$ is unbiased even when the true model is non-linear.	TRUE	OLS targets the best linear approximation; the moment condition $\mathbb{E}\{\mathbf{X}(Y - \mathbf{X}^\top \beta_0)\} = \mathbf{0}$ holds by construction.
3	The hat matrix $\mathbf{H}$ is idempotent: $\mathbf{H}^2 = \mathbf{H}$.	TRUE	$\mathbf{H}\mathbf{H} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top = \mathbf{H}$.
4	Raw residuals satisfy $\mathbb{E}(\mathbf{e}) = \mathbf{0}$ regardless of the true model.	FALSE	Requires the moment condition $\mathbb{E}\{\mathbf{X}^\top \varepsilon\} = \mathbf{0}$; not guaranteed under arbitrary misspecification.
5	High leverage h_{ii} implies the observation is influential.	FALSE	High leverage means the covariate is far from its mean; the observation is only influential if it also has a large residual. Cook's D_i combines both.
6	If $\hat{\beta}_1 = 1.6$ in logistic regression, the odds multiply by $e^{1.6} \approx 5.0$ per unit increase in x_1 .	TRUE	The logistic model gives $\log(\text{odds}) = \beta^\top \mathbf{x}$, so a unit increase in x_1 adds 1.6 to the log-odds.
7	LDA and logistic regression always produce the same decision boundary.	FALSE	Both produce linear boundaries but by different means. They coincide asymptotically under LDA assumptions, not in general.

1.2 Short Conceptual Questions

Q1. Logit link and inverse logit.

Answer: The logit link is $g(\pi) = \log\left(\frac{\pi}{1-\pi}\right)$. The inverse logit is $h(\eta) = \frac{e^\eta}{1+e^\eta} = \frac{1}{1+e^{-\eta}}$, the sigmoid function. It maps the real line to $(0, 1)$, is S-shaped, and equals 0.5 at $\eta = 0$.

Q2. OLS under misspecification.

Answer: Even if the true model is non-linear, OLS consistently estimates the best linear predictor $\beta_0 = \arg \min_{\beta} \mathbb{E}\|\mathbf{Y} - \mathbf{X}^\top \beta\|^2$. The moment condition $\mathbb{E}\{\mathbf{X}(Y - \mathbf{X}^\top \beta_0)\} = \mathbf{0}$ holds by construction, making $\hat{\beta}$ unbiased for β_0 . The statement confuses the estimator being sensible with the model being correctly specified.

Q3. Cook's distance $D_i = 0.8$.

Answer: Cook's D_i combines leverage h_{ii} (how unusual the covariate value is) and the squared standardised residual (how far the response is from the fit). A value of 0.8 is below the common threshold of 1 but not negligible. Recommended action: examine the observation, check for data-entry errors, and re-fit with and without the point to assess sensitivity. Do not automatically delete it.

1.3 Worked Example (Q4)

(a) How many parameters?

Answer: β_0 (intercept) + μ_2, μ_3 (two group dummies, group 1 as reference) + $\beta_1, \beta_2, \beta_3, \beta_4 = 7$ parameters.

(b) Linear in parameters vs. predictors.

Answer: Linear in parameters: yes — the model can be written $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ where $\boldsymbol{\beta}$ enters each term linearly. Linear in predictors: no, because of the z^2 and wz terms, which are non-linear functions of w and z . Being linear in parameters (required for OLS) is a strictly weaker condition than being linear in predictors.

(c) Normality of $\hat{\boldsymbol{\beta}}$.

Answer: Under $\boldsymbol{\varepsilon} \sim \mathcal{N}_n(\mathbf{0}, \sigma^2 \mathbf{I}_n)$, we have $\mathbf{Y} \sim \mathcal{N}_n(\mathbf{X}\boldsymbol{\beta}_0, \sigma^2 \mathbf{I}_n)$. Since $\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}$ is a linear transformation of $\mathbf{Y}$, and any linear transformation of a multivariate normal is multivariate normal, $\hat{\boldsymbol{\beta}}$ is normally distributed.

2 Block B: Model Evaluation

2.1 True / False

#	Statement	Answer	Notes
8	Accuracy is always the most appropriate metric for a binary classifier.	FALSE	Accuracy ignores class imbalance and weights all errors equally. Sensitivity, F1, or AUROC are preferred in many settings.
9	In K -fold CV each observation appears in the validation set exactly once.	TRUE	By construction, each observation belongs to exactly one fold.
10	LOOCV always has lower variance than 5-fold CV.	FALSE	LOOCV has low bias but can have <i>higher</i> variance because the n models are nearly identical (each training set differs by one observation).
11	The minimum test error coincides with the minimum training error.	FALSE	Training error decreases monotonically with complexity; test error is U-shaped. They rarely coincide.
12	The F1 score is the arithmetic mean of precision and recall.	FALSE	F1 is the <i>harmonic</i> mean: $F1 = 2PR/(P + R)$. It penalises imbalance between the two metrics more strongly.
13	In double descent, test error can decrease again after the interpolation threshold.	TRUE	In the overparameterised regime, implicit regularisation leads to a second decrease in test error.
14	Bootstrap resampling draws observations without replacement.	FALSE	Bootstrap samples with replacement — this is what allows repeated observations and mimics drawing new samples from the population.

2.2 Short Conceptual Questions

Q5. Confusion matrix calculations (TP=40, FP=10, FN=20, TN=130).

(a)

Answer:

- Accuracy = $(40 + 130)/200 = 0.85$
- Sensitivity = $40/(40 + 20) = 0.667$
- Specificity = $130/(130 + 10) = 0.929$
- Precision = $40/(40 + 10) = 0.80$
- F1 = $2 \times 0.80 \times 0.667 / (0.80 + 0.667) \approx 0.727$

(b)

Answer: Classifier 2 accuracy = $(55 + 110)/200 = 0.825$. Both have similar accuracy ($\approx 82\text{--}85\%$), but Classifier 2 has sensitivity = $55/60 \approx 0.917$ (far fewer missed cancers) at the cost of lower specificity = $110/140 \approx 0.786$. For cancer screening, missing a true case is far more costly than a false alarm, so Classifier 2 is clearly preferable. Accuracy alone hides this crucial difference.

Q6. Threshold and precision–recall trade-off.

Answer: Lowering the threshold labels more observations as positive. More true positives are captured (FN decreases), raising recall = $TP/(TP + FN)$. But more false positives are also introduced (FP increases), lowering precision = $TP/(TP + FP)$. At threshold = 0, everything is labelled positive: recall = 1 but precision equals the class base rate.

2.3 Worked Example (Q7)

(a) Why does larger K reduce CV bias?

Answer: Each model is trained on $n(1 - 1/K)$ observations. As K grows, training-set size approaches n , so models fitted inside CV more closely resemble the final model. The CV error estimate is therefore less pessimistically biased.

(b) Why not always use LOOCV?

Answer: LOOCV requires fitting n models — computationally expensive for large n or slow models. The n models are also highly correlated, which can increase variance. $K = 5$ or $K = 10$ gives a better practical balance.

(c) Double descent.

Answer: Stage 1 ($p < n$, underparameterised): test error decreases as the model gains capacity. Stage 2 (near $p \approx n$, interpolation threshold): test error peaks — the model barely interpolates the training data with an unstable, high-variance fit. Stage 3 ($p \gg n$, overparameterised): test error decreases again — very large models find minimum-norm solutions via implicit regularisation that happen to generalise well. The classical U-shape is only the first part of the full curve.

3 Block C: Regularisation and Bayesian Methods

3.1 True / False

#	Statement	Answer	Notes
15	Ridge shrinks all coefficients by the same multiplicative factor for orthonormal predictors.	TRUE	$\hat{\beta}_j^{\text{Ridge}} = \hat{\beta}_j^{LS} / (1 + \lambda)$: uniform proportional shrinkage.
16	Lasso can set coefficients exactly to zero; Ridge cannot.	TRUE	Soft thresholding produces exact zeros. Ridge scales but never zeros.
17	Adding $\lambda \mathbf{I}_p$ to $\mathbf{X}^\top \mathbf{X}$ guarantees invertibility for any $\lambda > 0$.	TRUE	All eigenvalues exceed $\lambda > 0$, so the matrix is strictly positive definite.
18	As $\lambda \rightarrow \infty$, $\hat{\beta}_{PLS} \rightarrow \hat{\beta}_{LS}$.	FALSE	As $\lambda \rightarrow \infty$, $\hat{\beta}_{PLS} \rightarrow \mathbf{0}$. As $\lambda \rightarrow 0$, $\hat{\beta}_{PLS} \rightarrow \hat{\beta}_{LS}$.
19	In the Bayesian framework, the prior is updated by the likelihood to produce the posterior.	TRUE	Bayes' theorem: $p(\theta y) \propto p(y \theta) p(\theta)$.
20	A conjugate prior produces a posterior from the same parametric family.	TRUE	Definition of conjugacy.
21	The MAP estimate under a Gaussian prior on β equals the Ridge estimator.	TRUE	Maximising the log-posterior under $\beta \sim \mathcal{N}(\mathbf{0}, \tau^2 \mathbf{I})$ is equivalent to minimising the Ridge criterion with $\lambda = \sigma^2 / \tau^2$.

3.2 Short Conceptual Questions

Q8. Geometry of Ridge vs Lasso.

Answer: The constrained optimum lies where the least-squares contour (an ellipse) first touches the constraint region. The Ridge region is a circle with a smooth boundary — the contact point is almost never on a coordinate axis, so no coefficient is set to zero. The Lasso region is a diamond with corners on the axes. Elliptical contours are more likely to first touch a corner, immediately setting one coefficient to zero. This is the geometric origin of Lasso sparsity. Both methods provide shrinkage; only Lasso provides sparsity.

Q9. Bayes' theorem terms.

Answer: $p(\theta | \mathbf{y}) = p(\mathbf{y} | \theta) p(\theta) / p(\mathbf{y}) \propto p(\mathbf{y} | \theta) p(\theta)$.

- $p(\theta)$: **prior** — belief about θ before observing data.
- $p(\mathbf{y} | \theta)$: **likelihood** — probability of the data given parameters.
- $p(\theta | \mathbf{y})$: **posterior** — updated belief after observing data.
- $p(\mathbf{y})$: **marginal likelihood** — normalising constant, independent of θ .

$p(\mathbf{y})$ is often an intractable high-dimensional integral and does not affect the shape of the posterior, so it is dropped for inference about θ .

3.3 Worked Example (Q10)

(a) Closed forms for orthonormal predictors.

Answer:

- Ridge: $\hat{\beta}_j(\lambda) = \hat{\beta}_{j,LS} / (1 + \lambda)$ — proportional shrinkage, never exactly zero.
- Lasso: $\hat{\beta}_j(\lambda) = \text{sign}(\hat{\beta}_{j,LS}) (|\hat{\beta}_{j,LS}| - \lambda/2)_+$ — soft thresholding, produces exact zeros for $|\hat{\beta}_{j,LS}| < \lambda/2$.

(b) Bias–variance trade-off for Ridge.

Answer: At $\lambda = 0$, Ridge equals OLS: unbiased, full variance. As λ increases, bias increases monotonically (coefficients are pulled toward zero) while variance decreases monotonically (better-conditioned inversion). $\text{MSE} = \text{Bias}^2 + \text{Variance}$ is U-shaped: there exists an optimal $\lambda^* > 0$ achieving lower MSE than OLS. Cross-validation finds λ^* in practice.

(c) Bayesian correspondence.

Answer: The posterior mean equals the Ridge estimator with $\lambda = \sigma^2/\tau^2$. Large τ^2 (diffuse prior) gives small λ (weak shrinkage, close to OLS). Small τ^2 (tight prior near zero) gives large λ (strong shrinkage toward zero).

4 Block D: Kernel and Exemplar Methods

4.1 True / False

#	Statement	Answer	Notes
22	In k -NN, smaller k gives lower bias and higher variance.	TRUE	$k = 1$ interpolates the training data exactly: zero training error, maximum test variance.
23	As $h \rightarrow 0$ in KDE, the estimator converges to a sum of Dirac deltas.	TRUE	Each kernel collapses to a point mass at the observation.
24	A larger bandwidth h always reduces bias in KDE.	FALSE	Larger h reduces variance but increases bias (over-smoothing).
25	The RBF kernel implicitly maps to a finite-dimensional feature space.	FALSE	The RBF kernel corresponds to an infinite-dimensional feature map.
26	Support vectors are the training observations closest to the decision hyperplane.	TRUE	They lie exactly on the margin boundary, $y_i f(\mathbf{x}_i) = 1$.

4.2 Short Conceptual Questions

Q11. k -NN calculations ($k = 5$, neighbours with labels $(+, +, -, +, -)$).

(a) Answer: Three of five neighbours are class $+$, so $\hat{P}(Y = + | \mathbf{x}) = 3/5 = 0.60$. Since $0.60 > 0.5$, the predicted class is $+$.

(b) Responses $(3.2, 4.1, 1.0, 3.8, 2.5)$.

Answer: $\hat{f}(\mathbf{x}) = (3.2 + 4.1 + 1.0 + 3.8 + 2.5)/5 = 14.6/5 = 2.92$.

Q12. Kernel trick.

Answer: In the dual SVM formulation the solution depends on the data only through inner products $\langle \mathbf{x}_i, \mathbf{x}_j \rangle$. After a feature mapping ϕ , these become $\langle \phi(\mathbf{x}_i), \phi(\mathbf{x}_j) \rangle$. A Mercer kernel satisfies $K(\mathbf{x}, \mathbf{z}) = \langle \phi(\mathbf{x}), \phi(\mathbf{z}) \rangle$ for some ϕ . Substituting K for every inner product allows us to work implicitly in the (possibly infinite-dimensional) feature space, evaluating only one scalar per pair without ever computing $\phi(\mathbf{x})$ explicitly.

4.3 Worked Example

Q13. Valid kernel properties for the Gaussian kernel $K(u) = \frac{1}{\sqrt{2\pi}} e^{-u^2/2}$.

Answer: A valid kernel density must satisfy: (1) non-negativity $K(u) \geq 0$; (2) symmetry $K(-u) = K(u)$; (3) unit integral $\int K(u) du = 1$. For the Gaussian: (1) the exponential is always positive; (2) $(-u)^2 = u^2$ so $K(-u) = K(u)$; (3) K is the standard normal density which integrates to 1. All three properties hold.

Q14. GP hyperparameter effects.

(a) Increasing noise variance σ^2 : the GP no longer interpolates the data exactly; the posterior mean is smoother and confidence bands are wider, even near training points. This explicitly accounts for measurement noise.

(b) Very small length-scale (large `sigma` in `rbfdot`): only very nearby points are correlated; the posterior mean follows rapid changes and may over-fit. Very large length-scale (small `sigma`): distant points remain correlated, producing a smooth slowly-varying function that may under-fit.

5 Block E: Tree-Based Methods and Boosting

5.1 True / False

#	Statement	Answer	Notes
28	CART selects splits by minimising RSS (regression) or Gini impurity (classification).	TRUE	Misclassification error is less sensitive to class-proportion changes and is not preferred for split selection.
29	Bagging trains each tree on a different bootstrap sample.	TRUE	Definition of bagging.
30	Approximately 63% of observations appear at least once in a bootstrap sample.	TRUE	$P(\text{selected}) = 1 - (1 - 1/n)^n \rightarrow 1 - e^{-1} \approx 0.632$.
31	Random forests use all p predictors at each split; bagging uses a random subset.	FALSE	The opposite: random forests use a random subset of <code>mtry</code> predictors; standard bagging uses all p .
32	In boosting, successive trees are fitted to the residuals of the current ensemble.	TRUE	Residuals capture what remains unexplained by the current model.
33	In AdaBoost, a classifier with $\text{err}_m = 0.5$ receives weight $\alpha_m = 0$.	TRUE	$\alpha_m = \log(0.5/0.5) = \log(1) = 0$: a random classifier contributes nothing.

5.2 Short Conceptual Questions

Q15. Decision tree example.

(a) Answer: Root node: the split $\text{Years} < 4.5$. Internal node: the split $\text{Hits} < 117.5$ (reached only when $\text{Years} \geq 4.5$). Terminal nodes (leaves): R_1, R_2, R_3 .

(b) Answer: $\text{Years} = 6 \geq 4.5$ and $\text{Hits} = 130 \geq 117.5$, so the player falls into R_3 . Predicted log-salary = 6.74, so predicted salary = $e^{6.74} \approx \$847$.

Q16. Random forests and decorrelation.

Answer: In bagging, strong predictors dominate splits in nearly every tree, making the trees highly correlated (ρ large). In the variance formula $\rho\sigma^2 + (1 - \rho)\sigma^2/M$, the term $\rho\sigma^2$ does not vanish as M grows — it is irreducible. Random forests restrict each split to a random subset of `mtry` $\ll p$ predictors, preventing any one predictor from dominating. This reduces ρ , shrinks the irreducible term, and substantially lowers ensemble variance, at the cost of slightly higher per-tree bias.

5.3 Worked Example

Q17. AdaBoost weights for $\text{err}_1 = 0.1$, $\text{err}_2 = 0.3$, $\text{err}_3 = 0.45$.

Answer:

$$\alpha_1 = \log(0.9/0.1) = \log 9 \approx 2.20, \quad \alpha_2 = \log(0.7/0.3) \approx 0.85, \quad \alpha_3 = \log(0.55/0.45) \approx 0.20.$$

The most accurate classifier ($\text{err} = 0.1$) gets the highest weight. Classifier 3, barely above chance, contributes almost nothing. If $\text{err}_m = 0.5$ exactly, $\alpha_m = 0$ and the classifier is ignored entirely.

Q18. Gradient boosting and pseudo-residuals.

Answer: Gradient descent updates parameters in the direction of the negative gradient of the loss. In gradient boosting, the “parameters” are the function values $f(\mathbf{x}_i)$. Pseudo-residuals are the negative gradients of the loss with respect to the current predictions:

$$r_{im} = - \left[\frac{\partial L(y_i, f(\mathbf{x}_i))}{\partial f(\mathbf{x}_i)} \right]_{f=f_{m-1}}.$$

A new weak learner F_m is fitted to these pseudo-residuals (analogous to a gradient step direction) and the model is updated as $f_m = f_{m-1} + \nu F_m$. For squared loss, pseudo-residuals reduce to ordinary residuals $y_i - \hat{f}_{m-1}(\mathbf{x}_i)$.

6 Open Synthesis Questions

S1. Bandwidth and neighbourhood size across methods.

Answer: In k -NN, k is the bandwidth: small k gives a tight neighbourhood (low bias, high variance); large k averages widely (high bias, low variance). In KDE, h plays exactly the same role: small h produces a wiggly estimate; large h over-smooths. In kernel regression, h controls how widely responses are averaged. All three are local smoothers and face the same bias-variance trade-off; cross-validation selects the optimal bandwidth in each case.

S2. Credit default prediction ($\sim 2\%$ positive rate).

Answer: (i) Metrics: accuracy is misleading at 2% prevalence (predicting “no default” always achieves 98% accuracy). Prefer AUROC, sensitivity/precision trade-off, or F1. Weight false negatives more heavily if missed defaults are costly. (ii) Regularisation: with many correlated financial predictors, Ridge, Lasso, or Elastic Net are strongly advisable. (iii) Tree ensembles vs. logistic regression: gradient boosting and random forests often outperform logistic regression on tabular data with non-linear interactions. Use nested cross-validation to compare. Also weigh interpretability: logistic regression gives odds-ratio interpretations; tree models can be explained via SHAP values.

End of teacher sheet.